Imperial College London

Department of Earth Science and Engineering

MSc in Geo-Energy with Machine Learning and Data Science

Independent Research Project

Project Report

Developing Machine Learning Methods for Crystal Texture Analysis to Investigate Magmatic Evolution and the Onset of Explosive Volcanic Eruptions

by

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GitHub Repository: https://github.com/ese-ada-lovelace-2025/irp-ars223.git

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Abstract

[write at the end]

!!! – NOTE – Volcanological and geological interp. Section should just be held within the discussion section as part of interpretation of results.

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1 - Introduction

[Take partially from the project plan]

1.1 - Problem Outline

2 [Take partially from the project plan]

2.1 - Review of Existing Work

3 [Take partially from the project plan]

3.1 - Novel Components of Project

4 [Take partially from the project plan]

4.1 - Proposed Solution

5 [Take partially from the project plan]

2 - Methodology

2.1 - Dataset

[write in detail about aegina + non aegina + unified dataset/manifest – stats and how produced]

2.2 - Data Preprocessing

[describe the unified dataset process]

2.3 – Segmentation and Optimisation of PlagNet

[describe square images and background colour + image checks for anomalies with contact sheets etc.]

Investigate possible leakage

2.4 – Chosen Models

2.4.1 - Chosen Model 1

SimSiam

2.4.2 - Chosen Model 2

SimCLR

2.4.3 - Chosen Model 3

DeepCluster

2.4.4 - Chosen Model 4

swAv

DINOV2 towards end if any time left (somewhat incomparable to others)

2.5 – Volcanological and Geological Interpretation

[not sure in this yet – leave this until the end – can get into how the geology/crystal textures can effect results by certain features such as high or low brightness on cold CL or resorption textures not being segmented or picked up by plagnet model] [MOVE TO DISCUSSION]

---

3 - Results

3.1 - Model Performance

[compare and explain metrics and metric choices which are standard across each model then compare in a big chart with discussion and run issues/examples]

3.2 - Visualising Model Performance

[Some graphs on performance for each and some explanation]

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4 - Discussion

4.1 - Interpretation of Results

[blend of data insights and geological knowledge/research]

4.2 - Key Learnings from the Process

[make it a concise almost bullet pointed run down of key points]

4.3 – Importance and Transferability of Work

[do towards the end – might require some extra papers/research]

---

5 - Conclusions

5.1 - Suggestions for Improvement and Further Research Ideas

Summary of what was achieved…

Further ways to improve project…

Other directions for project to take…

5.2 - Project Summary

Bibliography

(Harvard Style, In-text Citations, Alphabetical List Ordering) – mix of references from project plan and additional new/relevant sources – total no. recommended/in exemplars ???

AI Acknowledgement

Services: ChatGPT, Claude

Producers: OpenAI, Anthropic

Links: https://chatgpt.com/, https://claude.ai/

How generative AI was used: ADD BEFORE SUBMISSION

Declaration regarding AI use: All work in this project plan has been produced and written solely by the author listed above despite the use of the Generative AI tool listed above.

NOTES – DELETE BEFORE SUBMISSION

Questions /To-do Regarding Report:

- Are there any section missing that need to be included?

Final Report – Limitations:

The Final Report, together with the code and defended in the viva, is worth **95%** of the IRP mark. It is a **mandatory** deliverable — failure to submit will result in an incomplete assessment. See the Schedule page for the deadline.

Formatting, title page requirements, word count rules, citation style, and submission requirements that apply to all written reports are described on the Written Reports page.

Limits

word limit	figure + table limit	max PDF file size
5,000	10	8 MB

What to include

Here, instead of imposing a specific set of sections/subsections, we provide the following guidance on what content your final report should include.

- **Abstract**. A concise summary (approximately less than 200 words) of the project's purpose, methodology, key results, and conclusions. The abstract should enable readers to quickly grasp the contributions and scope of your work.

- **Problem Description**: Clearly define the research question or commercial/engineering challenge. Be specific and ambitious, but avoid overpromising.

    - **Significance**: Explain why your work matters. What potential contribution does it make to the field? Could it have broader impact or practical applications?

    - **Review of Existing Work**: Provide a critical overview of relevant literature or existing solutions. Describe the current state of the art, key findings, and limitations. Explain how your work builds on or differs from what has been done before. What is the originality or novelty of your work?

    - **Objectives**: Outline clear, specific, and measurable objectives that guided your work.

- **Methodology**: Describe and justify the method(s). Why these methods were chosen? Are there alternative approaches you considered? Discuss any limitations of your chosen methodology and demonstrate critical thinking in evaluating your choices.

- **Results**: Present the key outcomes of your project, with appropriate figures and tables. Clearly explain and interpret what the results show. Don't just present the data — analyse it. What do the results mean in the context of your research question or problem? How do they compare to existing work? Are there any unexpected findings? Don't let the reader draw conclusions on their own — guide them through your analysis.

- **Discussion and Conclusions**: Discuss the implications of your results. What were the most challenging aspects of the project? What worked well and what did not? Analyse the strengths and limitations of your solution, and suggest possible next steps. Be formal and analytical — quantify your statements where possible and ensure that all conclusions are directly supported by your results. If relevant, briefly describe any planned future work related to your project.

Framing

The final report takes on the character of a research paper. At this stage, your task is to report your findings to the community and beyond. You want to convince your readers that your conclusions are well supported by evidence, that your methodology was sound, and that your analysis was thoughtful and rigorous. A good final report, like a good research paper, should clearly communicate what was done, why it matters, and what was learned — in a way that gives others confidence to cite your work, build on your results, or apply your insights in practice. It should demonstrate critical engagement with the problem, clarity of thought, and transparency in how results were obtained and interpreted.

Code – Rules:

The code required to **fully reproduce all results**, within reason, must be submitted to your IRP GitHub repository. Reproducibility is essential for ensuring that examiners and other researchers can verify your results and for demonstrating the integrity of your work. All code used to generate reported results, such as plots and tables, should be included.

You are free to organise your repository as you wish, as long as the pre-created files remain intact and it is clear where each component is, in what order scripts should be run, and how the results are generated. Do not submit this code in the deliverables/ directory. Follow best practices for sustainable and maintainable code, including appropriate documentation and clear naming. For more information, see GitHub repository.

If your project is external AND confidential, you do not need to submit your code to GitHub, but you must discuss in advance with both your supervisors how the code can be made available to your internal supervisor for assessment. For more details, refer to Confidentiality.

Only the code necessary to reproduce the results presented in your final report is expected to be submitted. We do not expect to see code drafts, non-functioning scripts, or irrelevant files in your main repository structure. That said, producing such code is a normal part of the development process, and you are expected to make regular commits throughout your project. Before the final deadline, you should tidy your repository by moving any "irrelevant" or exploratory code into a separate directory (e.g. dev/ or sandbox/) and include a note in your README clarifying that examiners do not need to review it.

Sharing data

Unless very small, **data files should normally not be included directly in your repository**. Depending on the licence of the raw dataset, you may instead provide links to the data in appropriate places (e.g. in a Jupyter notebook, the README, or documentation). If the licence does not allow you to share the raw data, do not copy it anywhere - you can assume that examiners will have access to it.

Intermediate data produced during your project (e.g. processed data, trained model weights, or cached results) do not need to be submitted. However, the code required to reproduce these intermediates from the raw data must be included.

If you choose to share intermediate data or model weights for convenience, you may upload them to OneDrive (or another suitable storage service) and provide links in your repository or report at the appropriate points. It is your responsibility to ensure your supervisors have access to these files.

Code extensions, patches, and pull requests

If your project extends the capabilities of or adds functionality to an existing codebase, you should think of your code as an extension or patch. You do not need to submit the original codebase - only the code you have developed. If it is absolutely necessary to include portions of the original codebase, ensure the license permits this and make it completely clear in your submitted code what constitutes your own work. Your repository should include all necessary information on how your extension can be installed or how the patch can be applied.

If the original codebase you are extending, improving, or fixing is open source and hosted on GitHub, we strongly encourage you to contribute to it via **pull requests** following established best practices. This approach provides clear documentation and a permanent record of your contributions. In your written reports and IRP repository, you can reference these pull requests via their GitHub URLs to showcase the code you developed. When pull requests provide a comprehensive record of your contributions, resubmitting the same code to your IRP repository is unnecessary, as the pull request history serves as sufficient evidence of your work on the original codebase.